



PDF Download
3660853.3660915.pdf
27 January 2026
Total Citations: 4
Total Downloads: 260

DL Latest updates: <https://dl.acm.org/doi/10.1145/3660853.3660915>

RESEARCH-ARTICLE

Is Artificial Intelligence able to Produce Content Appropriate for Education Level? A Review on ChatGPT and Gemini

MEHMET F KARACA, Tokat Gaziosmanpaşa University, Tokat, Turkey

Open Access Support provided by:

Tokat Gaziosmanpaşa University

Published: 25 May 2024

[Citation in BibTeX format](#)

AICCONF '24: Cognitive Models and
Artificial Intelligence Conference
May 25 - 26, 2024
#istanbul, Türkiye

Is Artificial Intelligence able to Produce Content Appropriate for Education Level? A Review on ChatGPT and Gemini

Mehmet F. Karaca

Tokat Gaziosmanpaşa University, Erbaa Faculty of Social Sciences and Humanities, Department of Management Information Systems, Erbaa, Tokat, Turkey
mehmetfatih.karaca@gop.edu.tr

ABSTRACT

This study examined 120 Turkish stories written for primary, secondary, high school, and undergraduate education levels by ChatGPT-3.5, ChatGPT-4, and Gemini1.5 Pro. The data was processed by software created with Natural Language Processing methods in mind. The general characteristics, quantitative characteristics, and readability of the stories were all reviewed within the scope of the study. Using the Ateşman and Bezirci-Yılmaz formulas, which are widely used for Turkish texts, the readability of the stories was calculated. As a result of the analysis, it was determined that AI is able to produce distinct stories, and it generates cohesive stories by utilizing subjects and themes that are suitable for a specific educational audience. When the average readabilities are taken into account, it has been found that ChatGPT-3.5 generates better stories suited for the education level based on the Ateşman formula and Gemini based on the Bezirci-Yılmaz formula, and the difficulty level of ChatGPT-3.5 stories rises in tandem with education level in both formulas. Also, the stories at the undergraduate level were found to be the hardest to read and with primary schools having the easiest readability. When the number of stories at readability levels is taken into account, it has been found that GPT-3.5 and ChatGPT-4 in the Ateşman formula and ChatGPT-3.5, ChatGPT-4, and Gemini in the Bezirci-Yılmaz formula generate appropriate stories for the education level; the levels range from easy to difficult as the education level increases. Additionally, it has been found that the number of stories included is gradually increasing. It was concluded that AIs produced stories above their educational level; while ChatGPT-3.5 and Gemini were more successful in story production, the Bezirci-Yılmaz formula was better in determining readability.

CCS CONCEPTS

• Information systems; • Information retrieval; • Computing methodologies; • Machine learning;

KEYWORDS

Artificial Intelligence, ChatGPT, Gemini, Readability, Natural Language Processing

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

AICCONF '24, May 25, 26, 2024, İstanbul, Türkiye

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1692-8/24/05

<https://doi.org/10.1145/3660853.3660915>

ACM Reference Format:

Mehmet F. Karaca. 2024. Is Artificial Intelligence able to Produce Content Appropriate for Education Level? A Review on ChatGPT and Gemini. In *Cognitive Models and Artificial Intelligence Conference (AICCONF '24)*, May 25, 26, 2024, İstanbul, Türkiye. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3660853.3660915>

1 INTRODUCTION

It was the debate over whether machines could think that gave rise to artificial intelligence (AI). AI is the definition of machines that, through training and extensive data analysis, simulate human intelligence. AI is, in fact, not a novel idea. The fact that chatbots have been made available to the public by various AI platforms in 2023, their ease of use and capacity to conduct spoken language transactions without the need for technical knowledge, their ability to respond as though you are speaking with a human, and their Natural Language Processing (NLP) capabilities have all contributed to the significant rise in popularity of AI.

Users are most familiar with and utilize ChatGPT [1] and Gemini [2] among other AI platforms. AI and chatbots have gained increased visibility in our lives since the release of the ChatGPT-3.5 model by OpenAI in 2023 and the Gemini 1.5 Pro version by Google in 2024. ChatGPT, an AI-powered chatbot, comes in two versions: ChatGPT-3.5, which is free, and ChatGPT-4, which is a paid version. Gemini, formerly Bard, was introduced as Gemini 1.0 in December 2023 and Gemini 1.5 Pro in February 2024.

AI is currently capable of a wide range of tasks, including writing stories, editing text for grammar and spelling, answering exam questions, translating, coding, and creating presentations. These are not the only things AI is capable of. Given its widespread application and ongoing development, it wouldn't be incorrect to predict that updates from AI will lead to an even greater diversity of transactions in the future.

Can AI consider a user's educational attainment when creating content? Or can AI create content that is appropriate for various educational levels? The degree of readability of the produced textual content can be used to assess its appropriateness for a specific educational level. One of the measurable aspects of texts is their readability. Using the information gleaned from the texts in formulas, the readability of the text is assessed. Readability, which provides information about whether a text is easy or difficult, refers to the ease or difficulty of reading texts. To increase text readability, it is necessary to use short and clear words and short sentences. Turkish texts can be readability assessed using the Ateşman [3], Bezirci-Yılmaz [4], and Çetinkaya-Uzun [5] formulas.

Table 1: Number of stories according to education levels

Education Level of the Story	ChatGPT-3.5	ChatGPT-4	Gemini
Primary School	10	10	10
Secondary School	10	10	10
High School	10	10	10
Undergraduate	10	10	10
Total	40	40	40

1.1 Purpose of the study

The purpose of this study was to determine whether AI, which has gained a lot of popularity lately, can produce texts that are suitable for a given educational level. First of all, stories appropriate for varying educational levels were requested from the AI platforms ChatGPT-3.5, ChatGPT-4, and Gemini (Gemini 1.5 Pro). The readability of the stories generated by AI was then determined using the readability formulas developed by Ateşman and Bezirci-Yılmaz. The results obtained allowed for a comparative analysis of the AI's performances in content production, which has grown to be a popular data source for a wide range of topics. The following questions were the focus of this study, which will be the first of its kind in the literature:

- Is it possible for AI to create stories?
- What quantitative traits do stories produced by AI possess?
- What is the readability of AI-generated stories?
- Can AI create stories that fit different educational levels?
- Which AI is more adept at crafting stories that fit the education level?
- What are the number of stories at formula levels according to education levels?

2 METHOD

This study examined the quantitative features and readability of the stories generated by AI platforms through a descriptive survey carried out using quantitative research methods.

2.1 Dataset

Table 1 provides details about the dataset that was used for this study. Ten stories from each of the four educational levels, forty stories from each AI, and 120 Turkish stories make up the dataset. To compare AI performances, the same number of stories from three different platforms were used.

2.2 Data collection and processing

The collection of data and the written expression of sequence numbers were carried out manually, while other operations were conducted through software created using NLP techniques. With the software, the number of sentences and words in the stories, as well as the number of syllables and letters of the words, and the rate of unique words were determined.

A chat was started in ChatGPT-3.5, ChatGPT-4 and Gemini 1.5 Pro to gather data and create stories appropriate for students in primary, secondary, high school, and undergraduate programs. "Can you create 10 stories suitable for primary school education

level?" was the first question posed to three AIs. When presented in this way though, it was noted that all three generated extremely brief stories, each consisting of roughly three sentences. When the question, "Can you create a story suitable for primary school education level?" was asked afterward, they created longer stories. Hence, on March 15, 2024, the following questions were asked ten times to provide AI platforms with the ability to produce stories that are suitable for different educational levels:

- Can you create a story appropriate for the primary school education level?
- Can you create a story appropriate for the secondary school education level?
- Can you create a story appropriate for the high school education level?
- Can you create a story appropriate for the undergraduate education level?

The data processing step was initiated following the data collection phase. Initially, the punctuation marks (. ? ! . . .) indicating sentences were counted to ascertain the total number of sentences in the stories. The stories were then split up based on the space character, and the word count was determined by counting how many parts were created. The length of a word was used to calculate the number of letters, and the number of syllables was determined by counting the vowels. The stories were also processed using Zemberek, the Turkish NLP library, to identify unique words, and the stems and roots of the words that made up the stories were identified [6]. $UniqueWordNumber/TotalWordNumber$ was used to generate the unique word ratio, which was used to show the percentage of various words in the stories. The high percentage of unique words suggests that the stories' word variation.

To be used and processed in the following stages, the acquired data was entered into the database. The readability of the stories, the levels that correspond to the readability number, the values of the variables to be used in the readability formulas, and the quantitative data of the stories were all obtained by developed software.

2.3 Readability formulas

Readability is one of the quantitatively measurable characteristics of texts. Because of this characteristic, readability formulas are used to determine a text's level of difficulty or ease objectively. Three well-known and frequently used readability formulas for Turkish are Ateşman [3], Bezirci-Yılmaz [4], and Çetinkaya-Uzun [5].

Table 2 provides details on the Ateşman and Bezirci-Yılmaz readability formulas utilized in the present study. The formulas in the table are used to calculate readability using the variables derived

Table 2: Readability formulas

Readability formula	Calculation Formula	Readability Number (Value) Range / Readability Level
Ateşman	$= 198.825 - 40.175 \cdot X1 - 2.61 \cdot X2$ X1: Average word length by number of syllables X2: Average sentence length by word count	90-100 / Very Easy 70-89 / Easy 50-69 / Medium Difficulty 30-49 / Difficult 1-29 / Very Difficult
Bezirci-Yılmaz	$= \sqrt{OKS \cdot (H3 \cdot 0.84 + H4 \cdot 1.5 + H5 \cdot 3.5 + H6 \cdot 26.25)}$ OKS: Average sentence length by word count H3: Average number of 3-syllable words in a sentence H4: Average number of 4-syllable words in a sentence H5: Average number of 5-syllable words in a sentence H6: Average number of words of 6 or more syllables in a sentence	1-8 / Primary Education 9-12 / High School 13-16 / Undergraduate 16+ / Academic

from the texts. These variables are handled differently by each formula's coefficients. The Ateşman formula uses data on the average number of words in the sentence and the average number of syllables in the word. The Bezirci-Yılmaz formula examines and processes the average number of words in the sentence and the number of syllables in the word in more detail in terms of 3, 4, 5, 6, or more.

3 FINDINGS

In this section, the data obtained as a result of processing 120 Turkish story texts are presented. Observations on story production and quantitative data of the stories is first provided, followed by the average readability of the stories generated by AI platforms according to education level; lastly, the number of stories in the readability levels according to the formulas is delivered. Readability formulas do not directly take sentence count into account. For readability the average number of words in a sentence and the average number of syllables in a word are crucial factors. Other quantitative data were provided in addition to these, which are crucial for readability.

It has been noted that when AI platforms are asked to produce stories, they produce different stories each time. However, when asked to create collective stories such as "Can you create 10 stories suitable for primary school education level?", it was found that they typically produced short stories with three sentences on average. Additionally, it was discovered upon examination of the contents that each story had a plot and was constructed with logical integrity.

While stories at the primary school level largely focus on virtues like friendship and camaraderie, those at the secondary school level contain educational, instructive, and entertaining content related to nature, history, science, and personal development. High school stories consist of inspiring and encouraging adventures and fantasy elements related to science-fiction, technology, discovery, self-discovery and personal development. The stories at the undergraduate level contain intellectual and thought-provoking content about archaeology, history, art, literature, existence, philosophy, psychology, mathematics, biology, engineering, architecture, technology, science, and AI.

Table 3 displays the quantitative information that was gathered during the processing of the stories. The stories generated by Gemini were found to have an average length of sentences, words, and letters as well as an average number of letters in syllables that was higher than that of ChatGPT-3.5 and ChatGPT-4. It was discovered that there were fewer words in each sentence on average, fewer syllables and letters in each word on average, and a lower percentage of unique words. There was no consistency in the average number of sentences and unique word rate of stories between AI platforms or even within them. Conversely, the average word count and unspaced letter count of the stories produced by ChatGPT-3.5, ChatGPT-4, and Gemini (excluding ChatGPT-4), as well as the average word count of the sentences (excluding Gemini), and the average word count and syllable count of the words (excluding ChatGPT-4 and Gemini), all demonstrated an increase with the level of education.

Table 4 shows the average readability levels of stories produced at different education levels, as calculated using the Ateşman formula. Stories generated with ChatGPT-3.5 at the primary school level are easy, while those at the secondary school and high school levels are of medium difficulty, and undergraduate-level stories are difficult. On the other hand, stories produced with ChatGPT-4 at the primary school, secondary school, and high school levels are of medium difficulty, while undergraduate stories are difficult. The average readability of stories generated with ChatGPT-3.5 aligns well with the education level, becoming progressively more difficult with higher education levels. For stories produced with ChatGPT-4 and Gemini, the average readability is partially suitable for the education level, with the suitability being affected by the primary school level for ChatGPT-4 and the high school level for Gemini.

Table 5 displays the average readability values and levels of the stories based on the Bezirci-Yılmaz formula. It has been discovered that stories generated by ChatGPT-3.5 at the secondary school and undergraduate levels, ChatGPT-4 at the primary school, secondary school, high school, and undergraduate levels, and Gemini at the undergraduate level are all above the level. Generally, stories generated by AI platforms are above the level. Based on the results, it was concluded that the story difficulty level in ChatGPT-3.5 and Gemini increased in tandem with the education level. It has also

Table 3: Quantitative data obtained from stories

Quantitative Data	Education Level of the Story	ChatGPT-3.5	ChatGPT-4	Gemini
Average Number of Sentences of Stories	Primary School	22	22	22
	Secondary School	21	18	27
	High School	19	13	29
	Undergraduate	20	17	23
Average Word Count of Stories	Primary School	177	246	168
	Secondary School	201	229	230
	High School	213	178	244
	Undergraduate	242	239	269
Average Number of Letters in Stories (Unspaced)	Primary School	1091	1498	984
	Secondary School	1254	1470	1424
	High School	1342	1130	1467
	Undergraduate	1638	1617	1929
Average Word Number of Sentences	Primary School	8.06	11.29	7.76
	Secondary School	9.59	12.93	8.57
	High School	11.41	13.62	8.51
	Undergraduate	12.26	14.32	11.96
Average Number of Syllables of Words	Primary School	2.60	2.58	2.43
	Secondary School	2.70	2.77	2.67
	High School	2.71	2.75	2.64
	Undergraduate	2.93	2.95	3.05
Average Number of Letters in Words	Primary School	6.18	6.08	5.87
	Secondary School	6.23	6.43	6.18
	High School	6.29	6.33	6.02
	Undergraduate	6.78	6.76	7.17
Average Number of Letters in Syllables	Primary School	2.38	2.36	2.42
	Secondary School	2.31	2.32	2.31
	High School	2.32	2.30	2.29
	Undergraduate	2.31	2.29	2.35
Unique Word Rate	Primary School	0.28	0.29	0.29
	Secondary School	0.31	0.33	0.24
	High School	0.32	0.31	0.21
	Undergraduate	0.28	0.31	0.24

Table 4: Average readability numbers / levels of education levels according to Ateşman formula

Education Level of the Story	ChatGPT-3.5	ChatGPT-4	Gemini
Primary School	73.34 / Easy	65.55 / Medium Difficulty	80.96 / Easy
Secondary School	65.30 / Medium Difficulty	53.62 / Medium Difficulty	69.18 / Medium Difficulty
High School	60.02 / Medium Difficulty	52.84 / Medium Difficulty	70.70 / Easy
Undergraduate	49.07 / Difficult	42.93 / Difficult	44.96 / Difficult

been found that Gemini is the most effective AI platform for creating stories that fit the educational level; ChatGPT-3.5, the free version, generates stories that are better suited for the level than ChatGPT-4, the paid version.

Table 6 lists the number of stories at each level of education according to the Ateşman formula. Not a single story with difficult or very difficult at the primary school education level, very easy or very difficult at the secondary and high school education levels, very easy, easy, or very difficult at the undergraduate education level was found in any of the three AIs. Based on the data, it has

been ascertained that the majority of the stories are easy to medium difficult for primary school students, medium difficult for secondary and high school students, and difficult to medium-difficult for undergraduate students. It can be concluded that ChatGPT-3.5, ChatGPT-4, and partially Gemini generate stories that are appropriate for the education level based on the number of stories at readability levels.

Table 7 presents the distribution of stories at Bezirci-Yılmaz formula levels categorized by education levels. According to the Bezirci-Yılmaz formula mostly stories at the primary school education level are on primary education level, while the secondary

Table 5: Average readability values / levels of education levels according to the Bezirci-Yılmaz formula

Education Level of the Story	ChatGPT-3.5	ChatGPT-4	Gemini
Primary School	8.14 / Primary School	11.74 / High School	6.73 / Primary School
Secondary School	10.00 / High School	15.82 / Undergraduate	8.91 / Primary School
High School	12.07 / High School	15.79 / Undergraduate	9.11 / High School
Undergraduate	17.12 / Academic	20.18 / Academic	17.32 / Academic

Table 6: Number of stories at Ateşman formula levels according to education levels

Education Level of the Story	Readability Level	ChatGPT-3.5	ChatGPT-4	Gemini
Primary School	Very Easy	-	-	1
	Easy	7	1	9
	Medium Difficulty	3	9	-
	Difficult	-	-	-
	Very Difficult	-	-	-
Secondary School	Very Easy	-	-	-
	Easy	2	-	5
	Medium Difficulty	8	9	5
	Difficult	-	1	-
	Very Difficult	-	-	-
High School	Very Easy	-	-	-
	Easy	-	-	7
	Medium Difficulty	10	8	3
	Difficult	-	2	-
	Very Difficult	-	-	-
Undergraduate	Very Easy	-	-	-
	Easy	-	-	-
	Medium Difficulty	6	3	3
	Difficult	4	7	7
	Very Difficult	-	-	-

education level is both primary and high school levels. In addition, high school education level stories are around high school and undergraduate education level, and stories at the undergraduate education level are mostly academic level. The formula suggests that ChatGPT-3.5, ChatGPT-4, and Gemini generate stories that are appropriate for the educational level based on readability levels yet, the Ateşman formula may not yield the same results.

Different readability formulas use different variables with different coefficients; the formulas also differ in terms of level and number of levels. As a result, despite having been applied to the same stories, the data in Table 6 and Table 7 differ. One explanation for this could be that words with one or two syllables are not included in the Bezirci-Yılmaz formula, and words with three, four, five, or six syllables or more have varying effects on readability. Combining Table 6 and Table 7 reveals that the Bezirci-Yılmaz formula performs better in estimating the stories' educational levels.

4 CONCLUSION

A total of 120 Turkish stories created by ChatGPT-3.5, ChatGPT-4, and Gemini 1.5 Pro at the primary, secondary, high school, and undergraduate education levels were examined in this study. The general characteristics, quantitative features, and readability of the

stories were evaluated to figure out whether AI platforms generate stories that are appropriate for the educational level. To assess readability, the Ateşman and Bezirci-Yılmaz formulas were applied. NLP techniques were employed in the data processing process by the software created for the present study.

AI platforms have produced different stories each time. It has been noted that the stories use topics and themes that are appropriate for the readers' educational level, have content that will grab the interest of the intended audience, are not produced at random but rather are logically constructed, and have a plot. The average number of sentences and unique word rate of stories vary greatly amongst AI platforms, and even within the AI platform itself. However, it has also been realized that the average number of words and unspaced letters in the stories produced by ChatGPT-3.5 and Gemini, the average number of words in the sentences in the stories produced by ChatGPT-3.5 and ChatGPT-4, and the average number of syllables and letters in the stories produced by ChatGPT-3.5 all with the education level showed an increase in accordance with the education level; it was found that as the level increased, these values also increased.

The stories at the primary school education level were found to be the easiest to read, while the stories at the undergraduate

Table 7: Number of stories at Bezirci-Yılmaz formula levels according to education levels

Education Level of the Story	Readability Level	ChatGPT-3.5	ChatGPT-4	Gemini
Primary School	Primary School	8	1	10
	High School	2	6	-
	Undergraduate	-	2	-
	Academic	-	1	-
Secondary School	Primary School	4	-	6
	High School	5	1	4
	Undergraduate	1	6	-
	Academic	-	3	-
High School	Primary School	-	-	4
	High School	7	-	6
	Undergraduate	3	7	-
	Academic	-	3	-
Undergraduate	Primary School	-	-	-
	High School	1	-	1
	Undergraduate	3	-	2
	Academic	6	10	7

education level were found to be the most difficult in both formulas and three AI platforms, based on average readability. It has been discovered that there is little difference in readability between stories from primary, secondary, and high schools, but there is a significant difference between stories from undergraduate and high school education levels. Based on the Ateşman and the Bezirci-Yılmaz formula, it has been found that ChatGPT-3.5 and Gemini, respectively, generate stories that are more appropriate for the educational level. The average readability of the ChatGPT-3.5 and ChatGPT-4 stories is determined by the Ateşman formula, and the Bezirci-Yılmaz formula indicates that the average readability of the ChatGPT-3.5 and Gemini stories increases from easy to difficult based on the educational level.

Based on the number of stories at each formula level, it was observed that the Ateşman formula's GPT-3.5 and ChatGPT-4, as well as the Bezirci-Yılmaz formula's three AI platforms, generated stories appropriate for the school level. On the other hand, it was

found that the number of stories ranging from easy to difficult levels increased along with education level.

Based on all of these, it can be said that: AIs typically generate stories above the educational level, ChatGPT-3.5 and Gemini are more successful at producing stories, and the Bezirci-Yılmaz formula is more effective at determining readability.

REFERENCES

- [1] ChatGPT. 2024. "An artificial intelligence model", [Online]. Available: <https://chat.openai.com/>
- [2] Gemini. 2024. "An artificial intelligence model", [Online]. Available: <https://gemini.google.com/app>
- [3] Ateşman, E. 1997. Türkçede okunabilirliğin ölçülmesi. *Dil Dergisi*, (58): 71 74
- [4] Bezirci, B. & Yılmaz, A. E. 2010. Metinlerin okunabilirliğinin ölçülmesi üzerine bir yazılım kütüphanesi ve Türkçe için yeni bir okunabilirlik ölçütü. *Dokuz Eylül Üniversitesi Mühendislik Fakültesi Fen ve Mühendislik Dergisi*, 12(3): 49 62. <https://dergipark.org.tr/tr/pub/deumfmd/issue/40831/492667>
- [5] Çetinkaya, G. 2010. Türkçe metinlerin okunabilirlik düzeylerinin tanımlanması ve sınıflandırılması. (Published Doctoral Thesis). Ankara Üniversitesi, Sosyal Bilimler Enstitüsü
- [6] Zemberek. 2015, July 12. "An open source NLP library for Turkic languages", [Online]. Available: <http://code.google.com/p/nzemberek>